

Alberto Andrés Valdés González.

Degree: Mathematical Engineer.

Work position: ML-Engineer.

Mail: anvaldes@uc.cl/alberto.valdes.gonzalez.96@gmail.com

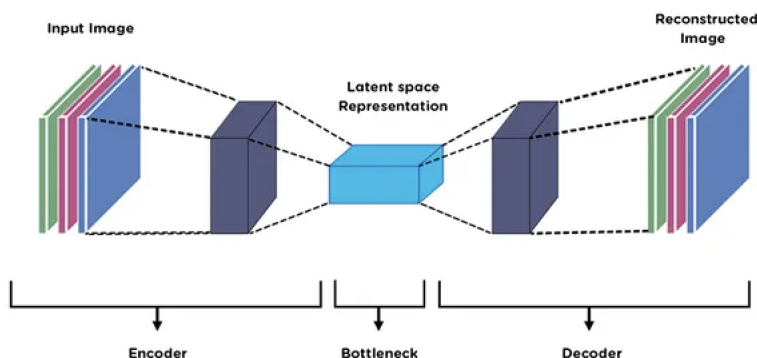
Location: Santiago, Chile.

Difussion Models

In machine learning, diffusion models use noise to generate data by adding random noise to data and then reversing the process. This process is similar to how molecules move from high to low concentration areas in physics.

Autoencoders

Auto encoders are neural networks that learn to reconstruct their inputs by encoding them into a lower-dimensional latent space and then decoding them back to the original space. The learning process is unsupervised, meaning that the model does not need labels to learn the representations.



However, auto encoders can suffer from instability and lack of robustness when learning representations. This is because they rely on the assumption that the data follows a certain distribution. Furthermore they can be sensitive to small changes in the data or the model parameters, which can lead to poor generalization and unreliable results.

Stable diffusion addresses these issues by introducing a regularization term to the learning process. This term encourages the model to preserve the distances between the data points in the latent space, even when the data or the model parameters change. This helps to improve the stability and robustness of the learned representations.

